

Benjamin Carver

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Summary

- PhD candidate in computer science with extensive experience designing, developing, and evaluating large, scalable, high-performance distributed systems across multiple cloud platforms.
- Adaptive learner and adept problem-solver with proficiency in a wide array of programming languages, development tools, frameworks, and cloud services.
- Established, facilitated, and lead multiple research collaborations with leading academic and industry partners.
- Motivated and driven to leverage strong academic and development background, along with substantial research expertise in distributed systems, cloud and serverless computing, and AI/ML, to help organizations build innovative, scalable, and high-performance infrastructure, products, and services.

Education

George Mason University, Fairfax, Virginia

- May 2025 *Computer Science*, Doctor of Philosophy, (GPA: 4.0).
May 2021 *Computer Science*, Master of Science, *Summa cum laude* (GPA: 4.0).
May 2020 *Computer Science*, Bachelor of Science, *Summa cum laude* (GPA: 4.0).

Employment

- Spring 2019–
Ongoing **Undergraduate/Graduate Research Assistant**, GEORGE MASON UNIVERSITY.
○ Served as the sole developer and tester of 3 complete, large-scale open-source distributed systems and as a co-developer of 2 other distributed applications of similar scale.
○ Experience creating and deploying distributed applications and services in cloud environments—including AWS, Google Cloud Platform, Microsoft Azure, and IBM Cloud—using Kubernetes and Docker (Compose/Swarm).
○ Published 4 academic papers in top-tier academic conferences, 3 of which were first-authorship publications, with 3 additional first-authorship papers under review or close to submission.
○ Established, lead, and facilitated multiple collaborations with leading academic and industry research labs.
- May–August
2022 **Microsoft Research Intern**, MICROSOFT RESEARCH LAB - REDMOND.
○ Team member responsible for designing, developing, and evaluating a new microsecond-latency FaaS platform.
○ Improved API usability and reduced overheads by 25% by re-implementing client-facing interfaces as CPython extension modules, enabling multi-threading support and improving client application performance.
- June
2016–July
2020 **Software Developer**, BRTRC FEDERAL SOLUTIONS.
○ Optimized a .NET-based Geographic Information Systems (GIS) application, reducing application load time by 5× and application memory footprint by 70%, significantly improving usability and the overall user-experience.
○ Implemented system for manipulating and visualizing realistic, user-defined terrain maps and custom 3D models.

Ongoing Research

Jupyter Notebooks-as-a-Service (NaaS) Platform

- Responsible for the design, implementation, and evaluation of a novel, large-scale, fully functional Jupyter Notebooks-as-a-Service platform with full support for GPU-based workloads like deep learning training.
- Developed novel design that enables support for new classes of AI/ML applications and workloads.
- Enabled detailed, real-time metric and log collection, visualization, and monitoring by integrating platform with multiple industry-standard frameworks and technologies—including Prometheus, Promtail, Loki and Grafana.
- Created robust, detailed simulation software to evaluate pricing/cost, scheduling, fault tolerance, and scaling algorithms and policies of the Jupyter NaaS platform at extremely large scales.

A Microsecond-Scale Functions-as-a-Service (FaaS) Platform

- Team member responsible for designing, implementing, and evaluating a novel FaaS platform created to support modern scientific and data-analytics-based workloads.
- Integrated support for latest networking and virtualization technologies to enable orders-of-magnitude reduction in overhead incurred by the serverless platform.

Enhancing FaaS-Based Storage Systems for Privacy & Security

- Tasked with designing a secure FaaS-based storage system that incorporates techniques from cryptography, network security, and database systems to provide strong security guarantees.

Publications

- [ASPLOS '23] **Benjamin Carver, Runzhou Han, Jingyuan Zhang, Mai Zheng, Yue Cheng**, *λ FS: A Scalable and Elastic Distributed File System Metadata Service using Serverless Functions*, ACM International Conference on Architectural Support for Programming Languages and Operating Systems.
- Designed, implemented, and evaluated λ FS, a first-of-its-kind serverless metadata service for large-scale distributed file systems like Hadoop Distributed File System (HDFS).
 - Designed and implemented elastic auto-scaling mechanisms, efficient client libraries, and a novel FaaS-based metadata cache to provide $4.13\times$ higher throughput, 90.40% lower latency, and 85.99% lower user-side cost compared to state-of-the-art baselines for real-world industrial workloads.
- [VLDB '23] **Ao Wang, Jingyuan Zhang, Xiaolong Ma, Benjamin Carver, Ali Anwar, Lukas Rupperecht, Dimitrios Skourtis, Vasily Tarasov, Feng Yan, Yue Cheng**, *InfiniStore: Elastic Serverless Cloud Storage*, International Conference on Very Large Databases.
- Team member assisting in the testing and evaluation of INFINISTORE, a first-of-its-kind serverless-function-based data storage service that achieves 97.24% user-side cost savings compared to AWS ElastiCache.
 - Implemented a lightweight, microservice-based workload driver in Golang to execute large-scale, customizable MapReduce workloads as part of the evaluation of INFINISTORE and its predecessor, INFINICACHE.
- [SOCC '20] **Benjamin Carver, Jingyuan Zhang, Ao Wang, Ali Anwar, Panruo Wu, Yue Cheng**, *Wukong: A Scalable and Locality-Enhanced Framework for Serverless Parallel Computing*, ACM Symposium on Cloud Computing (AR: $35/143 = 24.4\%$).
- Designed, implemented, and evaluated WUKONG, a fully-featured serverless-computing-based execution engine created atop AWS Lambda and AWS Fargate.
 - Developed novel scheduling algorithms that leverage serverless functions to execute large-scale parallel computing jobs up to $68.17\times$ faster while reducing user-side cost by 92.96% compared to state-of-the-art baselines.
- [PDSW '19] **Benjamin Carver, Jingyuan Zhang, Ao Wang, Yue Cheng**, *In Search of a Fast and Efficient Serverless DAG Engine*, International Parallel Data Systems Workshop.
- Created and evaluated system prototype that uses AWS Lambda to improve the performance of real-world data analytics, linear algebra, and machine learning workloads by up to $3.1\times$ compared to a non-serverless baseline.

Skills

Languages	Python, Golang, Java, C/C++, C#, JavaScript/TypeScript, CSS.
Development Tools	VS Code, Visual Studio, JetBrains IDEs (IntelliJ/GoLand/WebStorm), Linux, Microsoft Office, Git (GitHub/GitLab/BitBucket), Jupyter Notebooks.
Frameworks & Services	Kubernetes, Docker, Docker Compose, Docker Swarm, Prometheus, Grafana, Loki, Promtail, HDFS, Dask, Pandas, NumPy, SciPy, PyTorch, JupyterLab, React, ZooKeeper, Redis, MySQL, Ansible.
Cloud	Amazon Web Services (AWS) EC2, VPC, Lambda, S3, IAM, CloudWatch, Fargate, ECS, EKS. Google Cloud Platform (GCP) Compute Engine, Google Kubernetes Engine (GKE), Cloud Functions. Microsoft Azure Cloud, IBM Cloud

Honors & Awards

August 2023	VLDB2023 NSF Travel Grant.
May 2021	George Mason University Presidential Scholarship Award. Renewed annually for 4 years.
May 2021	Distinguished Academic Achievement Award.
May 2020	Distinguished Undergraduate Research Award.
May 2020	Distinguished Academic Achievement Award.

Relevant Academic Coursework

Systems	Operating Systems, Distributed Systems, Database Systems, Concurrent and Distributed Systems.
AI/ML	Large-Scale Optimization for Deep Learning, Deep Learning, Advanced Artificial Intelligence, Artificial Intelligence, Machine Learning, Data Mining.
Core	Data Structures, Computer Architecture, C Programming, Compilers, Object Oriented Software, Software Engineering, Algorithm Analysis, Graph Algorithms, Mobile App Development, Computer Networks.